

```
In [1]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

In [2]: calendar = pd.read_csv("C:\\Users\\rauty\\Downloads\\Retail-Business-Analytics\\data\\MavenMarket_Calendar.csv")
customers = pd.read_csv("C:\\Users\\rauty\\Downloads\\Retail-Business-Analytics\\data\\MavenMarket_Customers.csv")
products = pd.read_csv("C:\\Users\\rauty\\Downloads\\Retail-Business-Analytics\\data\\MavenMarket_Products.csv")
regions = pd.read_csv("C:\\Users\\rauty\\Downloads\\Retail-Business-Analytics\\data\\MavenMarket_Regions.csv")
returns = pd.read_csv("C:\\Users\\rauty\\Downloads\\Retail-Business-Analytics\\data\\MavenMarket>Returns_1997-1998.csv")
stores = pd.read_csv("C:\\Users\\rauty\\Downloads\\Retail-Business-Analytics\\data\\MavenMarket_Stores.csv")
transactions_1997 = pd.read_csv("C:\\Users\\rauty\\Downloads\\Retail-Business-Analytics\\data\\MavenMarket_Transactions_1997.csv")
transactions_1998 = pd.read_csv("C:\\Users\\rauty\\Downloads\\Retail-Business-Analytics\\data\\MavenMarket_Transactions_1998.csv")

In [3]: transactions = pd.concat(
    [transactions_1997, transactions_1998],
    ignore_index=True
)

In [4]: print(transactions.shape)
print(products.shape)
print(customers.shape)
print(stores.shape)
print(returns.shape)

(269720, 6)
(1560, 9)
(10281, 20)
(24, 13)
(7087, 4)

In [5]: #Merge into One Master Dataset

In [6]: master = transactions.merge(
    products,
    on="product_id",
    how="left"
)
```

```
master.head()
```

```
Out[6]:
```

	transaction_date	stock_date	product_id	customer_id	store_id	quantity	product_brand	product_name	product_sku	product_retail_price
0	1/1/1997	12/31/1996	869	3449	6	5	Nationeel	Nationeel Grape Fruit Roll	52382137179	1.5
1	1/1/1997	12/31/1996	1472	3449	6	3	Fort West	Fort West Fudge Cookies	37276054024	1.5
2	1/1/1997	12/28/1996	76	3449	6	4	Red Spade	Red Spade Sliced Chicken	62054644227	1.5
3	1/1/1997	12/26/1996	320	3449	6	3	Excellent	Excellent Cranberry Juice	36570182442	1.5
4	1/1/1997	12/25/1996	4	3449	6	4	Washington	Washington Cream Soda	64412155747	1.5

```
In [7]: master = master.merge(
        customers,
        on="customer_id",
        how="left"
    )
```

```
In [8]: master = master.merge(
        stores,
        on="store_id",
        how="left"
    )
```

```
In [9]: master = master.merge(
        regions,
        on="region_id",
        how="left"
    )
```

```
In [10]: master["Sales"] = master["quantity"] * master["product_retail_price"]
```

```
master["Profit"] = (  
    master["product_retail_price"] - master["product_cost"]  
) * master["quantity"]
```

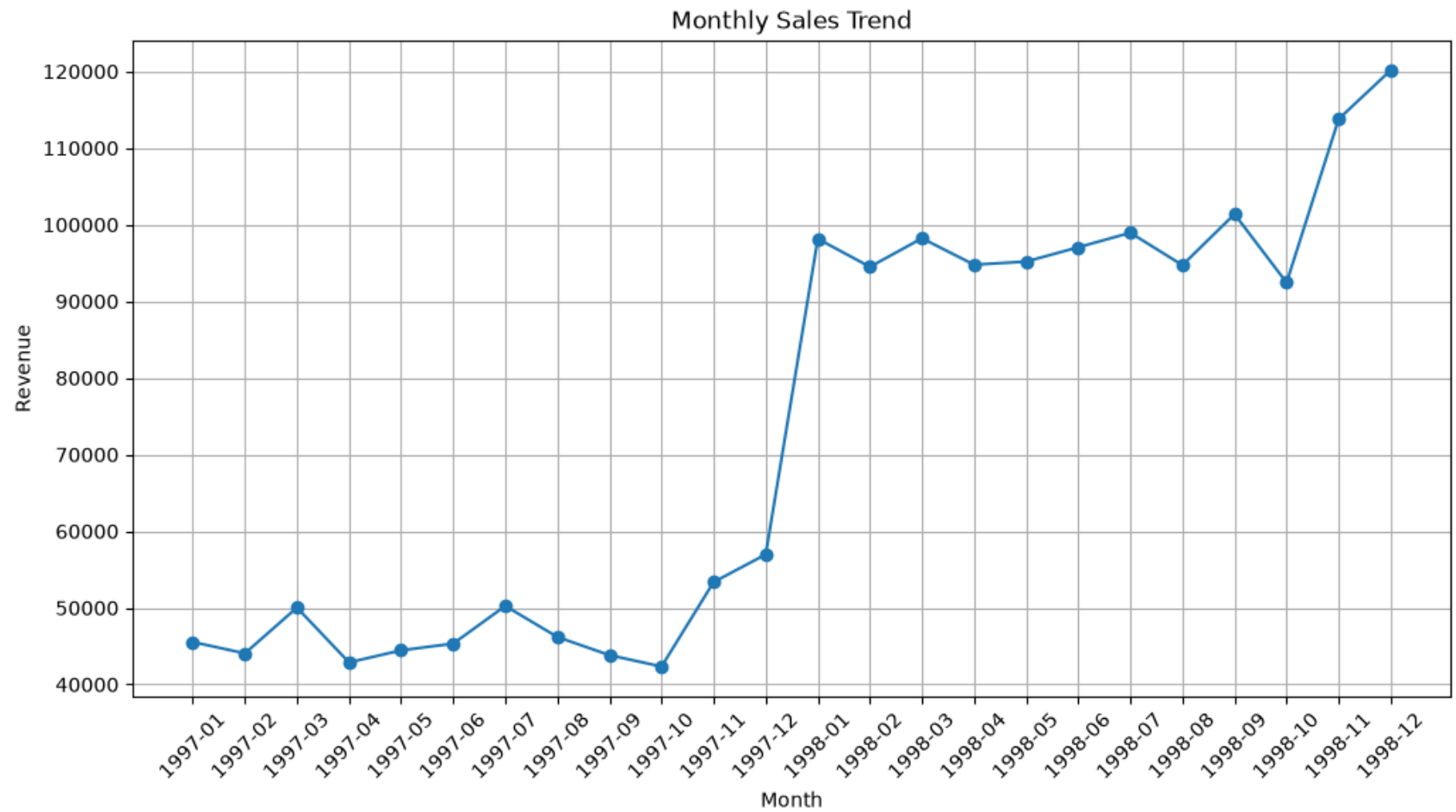
```
In [11]: master["transaction_date"] = pd.to_datetime(master["transaction_date"])
```

```
In [12]: master.shape
```

```
Out[12]: (269720, 49)
```

```
In [13]: ##Sales Trend Analysis
```

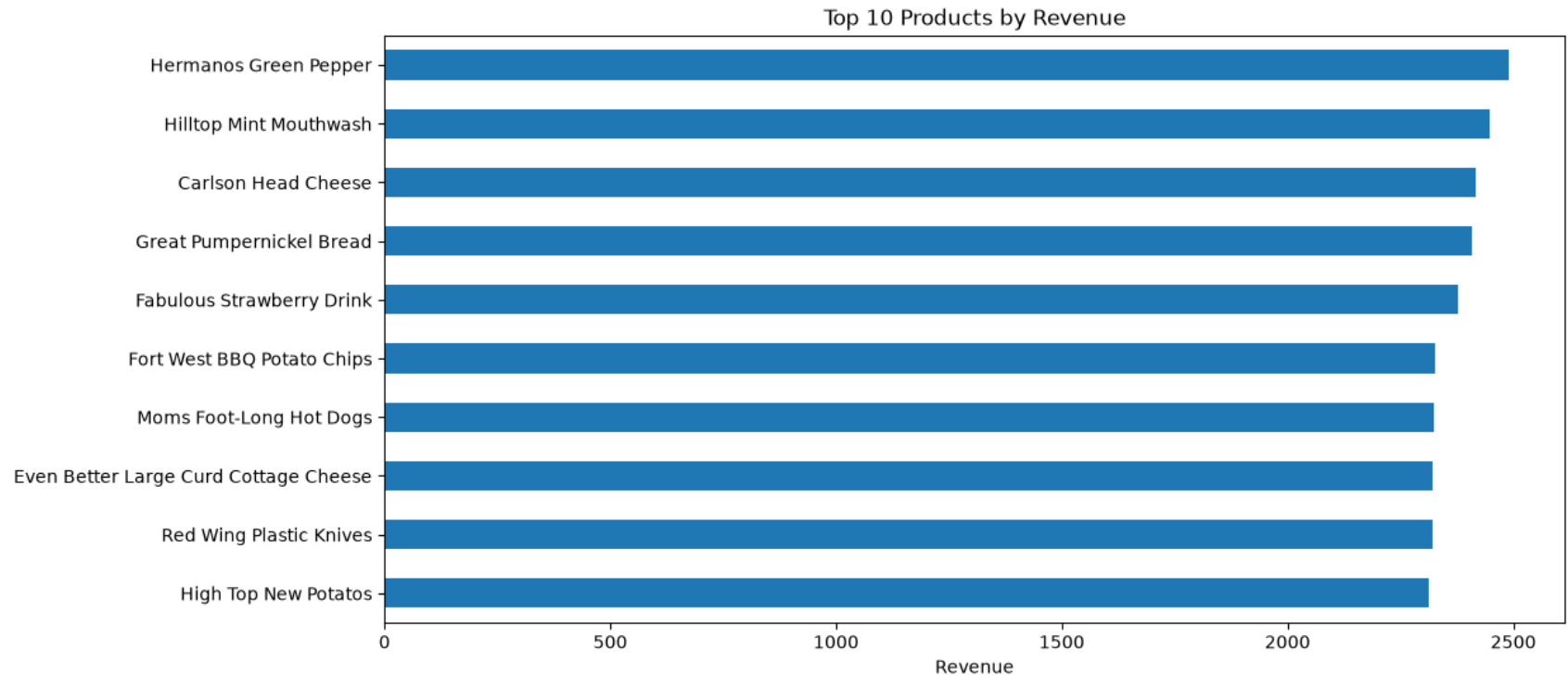
```
In [14]: master['transaction_date'] = pd.to_datetime(master['transaction_date'])  
  
monthly_sales = (  
    master.groupby(master['transaction_date'].dt.to_period('M'))['Sales']  
    .sum()  
    .reset_index()  
)  
  
monthly_sales['transaction_date'] = monthly_sales['transaction_date'].astype(str)  
  
plt.figure(figsize=(12,6))  
plt.plot(monthly_sales['transaction_date'], monthly_sales['Sales'], marker='o')  
plt.xticks(rotation=45)  
plt.title("Monthly Sales Trend")  
plt.xlabel("Month")  
plt.ylabel("Revenue")  
plt.grid(True)  
plt.show()
```



In [15]: `##Top 10 Products`

```
In [16]: top_products = (  
    master.groupby('product_name')['Sales']  
        .sum()  
        .sort_values(ascending=False)  
        .head(10)  
    )  
  
    plt.figure(figsize=(12,6))  
    top_products.sort_values().plot(kind='barh')  
    plt.title("Top 10 Products by Revenue")
```

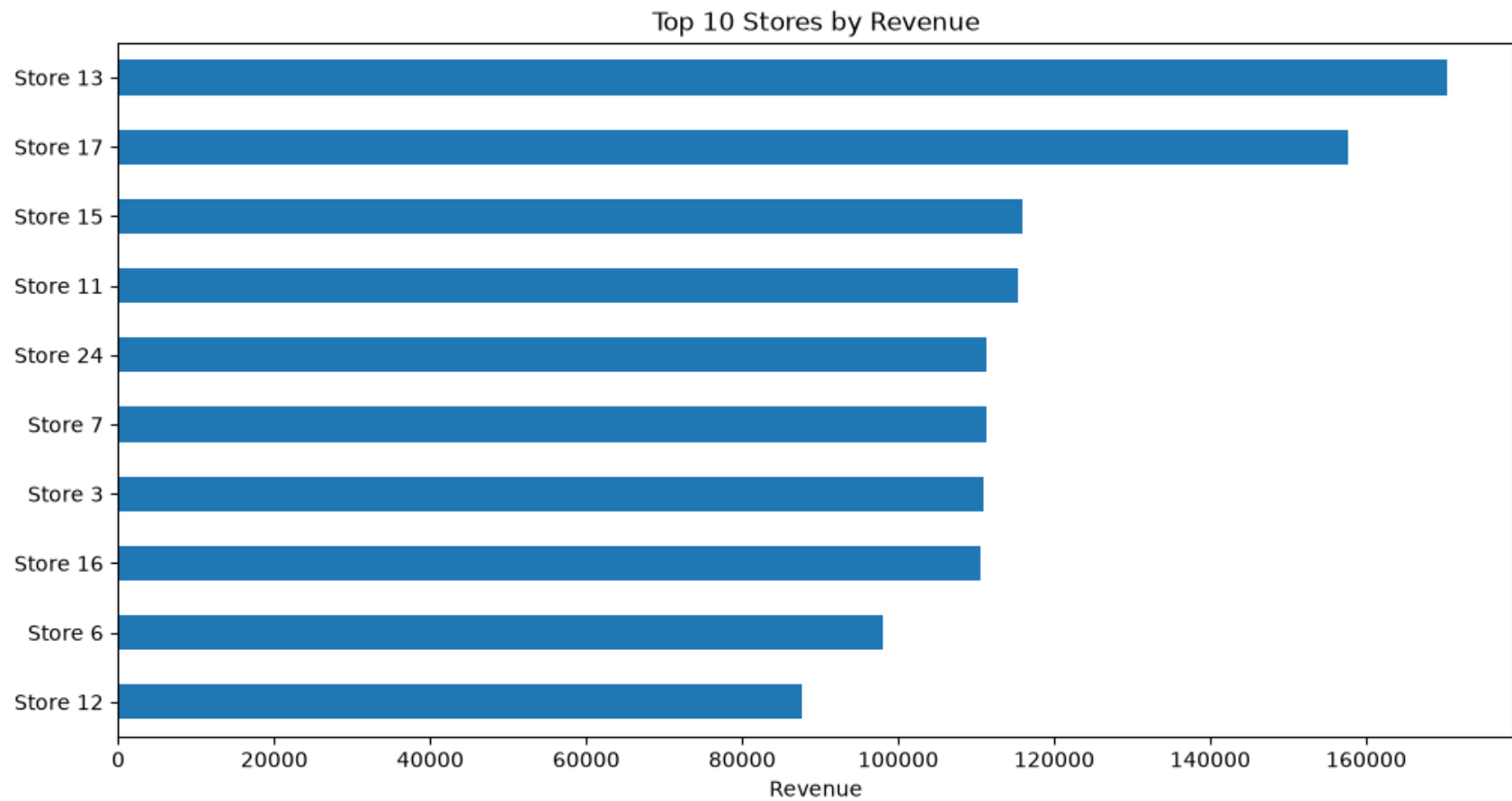
```
plt.xlabel("Revenue")
plt.ylabel("")
plt.show()
```



```
In [17]: ##Top 10 Stores
```

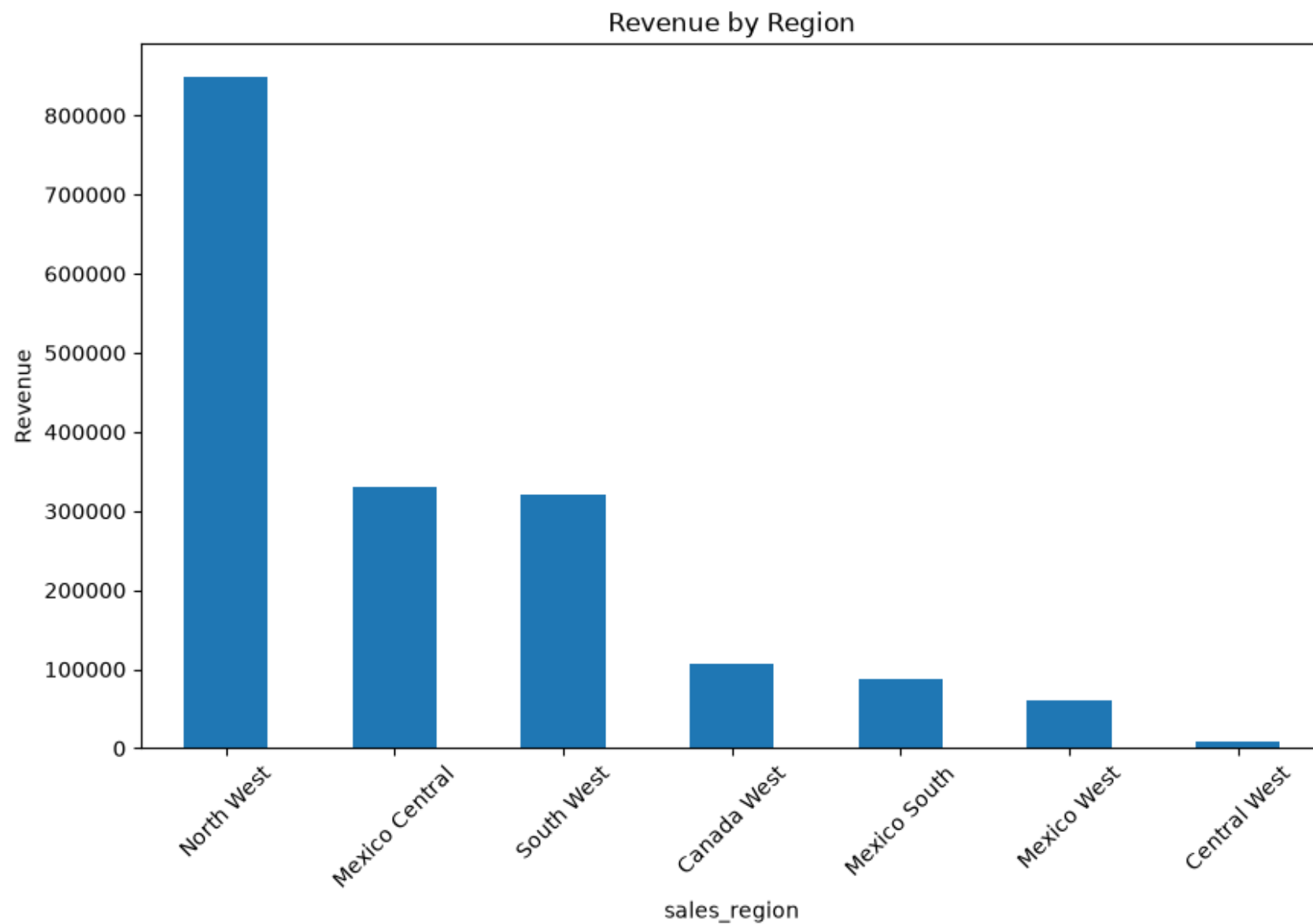
```
In [18]: top_stores = (
    master.groupby('store_name')['Sales']
        .sum()
        .sort_values(ascending=False)
        .head(10)
    )

plt.figure(figsize=(12,6))
top_stores.sort_values().plot(kind='barh')
plt.title("Top 10 Stores by Revenue")
plt.xlabel("Revenue")
plt.ylabel("")
plt.show()
```



In [19]: `# Revenue by Region`

```
In [20]: region_sales = (  
    master.groupby('sales_region')['Sales']  
        .sum()  
        .sort_values(ascending=False)  
    )  
  
    plt.figure(figsize=(10,6))  
    region_sales.plot(kind='bar')  
    plt.title("Revenue by Region")  
    plt.ylabel("Revenue")  
    plt.xticks(rotation=45)  
    plt.show()
```

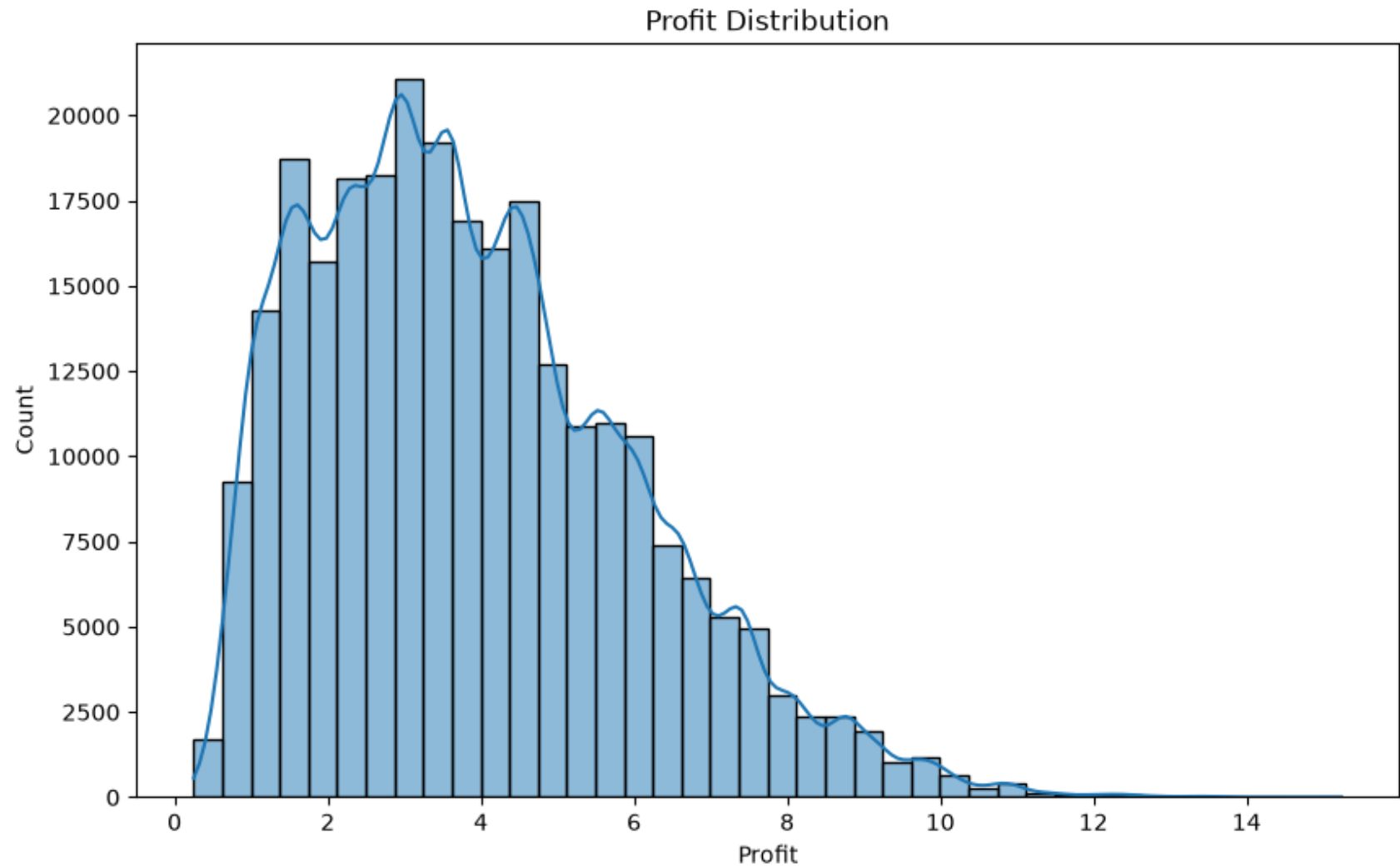


```
In [21]: # Profit Distribution
```

```
In [22]: plt.figure(figsize=(10,6))  
sns.histplot(master['Profit'], bins=40, kde=True)
```



```
plt.title("Profit Distribution")  
plt.xlabel("Profit")  
plt.show()
```



```
In [23]: # Sales vs Profit
```

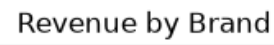
```
In [24]: plt.figure(figsize=(10,6))  
sns.scatterplot(  
    data=master,
```

```
x='Sales',  
y='Profit'  
)  
  
plt.title("Sales vs Profit")  
plt.show()
```



```
In [25]: # Revenue by Brand
```

```
In [26]: brand_sales = (  
    master.groupby('product_brand')['Sales']  
        .sum()  
        .sort_values(ascending=False)  
    )  
  
    plt.figure(figsize=(12,6))  
    brand_sales.plot(kind='bar')  
    plt.title("Revenue by Brand")  
    plt.ylabel("Revenue")  
    plt.xticks(rotation=45)  
    plt.show()
```



```
In [27]: # Interactive Sales Trend (Plotly)
```

```
In [28]: fig = px.line(
            monthly_sales,
            x='transaction_date',
            y='Sales',
            title='Monthly Sales Trend'
        )

fig.show()
```

```
In [29]: # Business Analytics
# Pareto Analysis (80/20 Rule)
# High Revenue but Low Margin Products
# Region Performance Matrix
# Store Performance
# Customer Segmentation
# Return Analysis
```

```
In [30]: # Pareto Analysis (80/20 Rule)
product_sales = (
    master.groupby("product_name")["Sales"]
        .sum()
        .sort_values(ascending=False)
        .reset_index()
)

product_sales["Revenue_%"] = (
    product_sales["Sales"] /
    product_sales["Sales"].sum()
) * 100

product_sales["Cumulative_%"] = (
    product_sales["Revenue_%"].cumsum()
)

product_sales.head()
```

```
Out[30]:
```

	product_name	Sales	Revenue_%	Cumulative_%
0	Hermanos Green Pepper	2489.70	0.141096	0.141096
1	Hilltop Mint Mouthwash	2447.12	0.138683	0.279778
2	Carlson Head Cheese	2417.73	0.137017	0.416795
3	Great Pumpernickel Bread	2408.56	0.136497	0.553293
4	Fabulous Strawberry Drink	2377.90	0.134760	0.688053

```
In [31]: plt.figure(figsize=(14,6))
```

```
plt.bar(
    product_sales["product_name"],
    product_sales["Revenue_%"]
)

plt.plot(
    product_sales["Cumulative_%"],
    color="red",
    linewidth=2
)

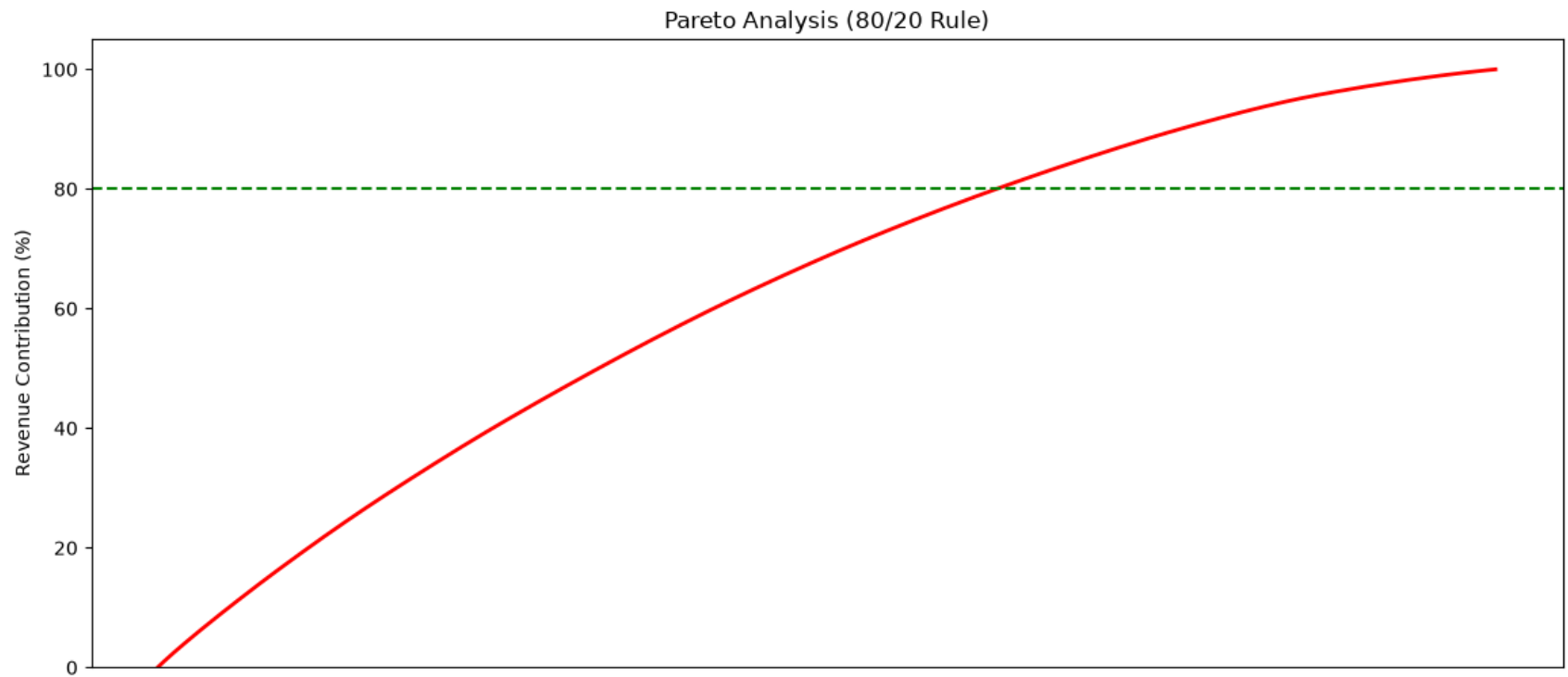
plt.axhline(80, color="green", linestyle="--")

plt.xticks([], [])

plt.title("Pareto Analysis (80/20 Rule)")

plt.ylabel("Revenue Contribution (%)")

plt.show()
```



```
In [32]: # High Revenue but Low Margin Products
product_analysis = (
    master.groupby("product_name")
    .agg(
        Revenue=("Sales", "sum"),
        Profit=("Profit", "sum")
    )
)

product_analysis["Margin"] = (
    product_analysis["Profit"] /
    product_analysis["Revenue"]
) * 100

product_analysis.sort_values(
    "Revenue",
    ascending=False
).head(20)
```

Out[32]:

	Revenue	Profit	Margin
product_name			
Hermanos Green Pepper	2489.70	1670.55	67.098446
Hilltop Mint Mouthwash	2447.12	1493.96	61.049724
Carlson Head Cheese	2417.73	1254.54	51.889169
Great Pumpernickel Bread	2408.56	1326.64	55.080214
Fabulous Strawberry Drink	2377.90	1474.90	62.025316
Fort West BBQ Potato Chips	2328.00	1326.00	56.958763
Moms Foot-Long Hot Dogs	2325.40	1183.84	50.909091
Even Better Large Curd Cottage Cheese	2322.60	1275.96	54.936709
Red Wing Plastic Knives	2321.25	1343.23	57.866667
High Top New Potatos	2313.85	1364.27	58.961039
Better Chicken Soup	2292.48	1307.52	57.035176
Just Right Rice Soup	2286.08	1258.56	55.053191
Tell Tale Cantelope	2284.36	1207.96	52.879581
Hermanos Lemons	2277.32	1340.96	58.883249
Modell Rye Bread	2275.47	1204.32	52.926209
Horatio Frosted Cookies	2262.00	1287.60	56.923077
Big Time Beef TV Dinner	2256.07	1309.79	58.056266
Horatio Fudge Cookies	2218.62	1130.94	50.974930
Tell Tale Fresh Lima Beans	2212.66	1347.14	60.883281
Super Extra Chunky Peanut Butter	2211.82	1484.73	67.127072

In [33]: # Region Performance Matrix

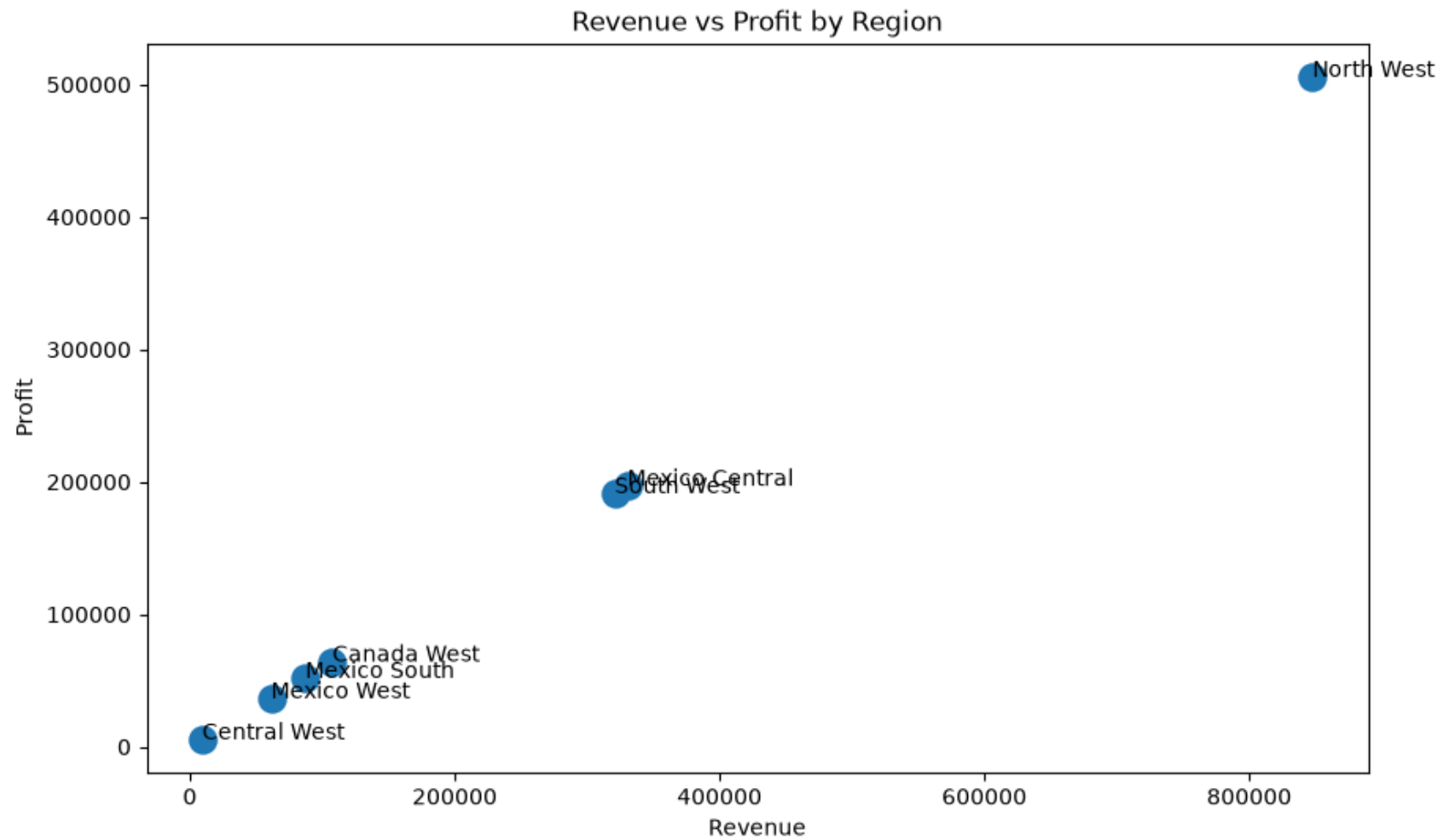

```
region_analysis = (  
    master.groupby("sales_region")  
    .agg(  
        Revenue=("Sales", "sum"),  
        Profit=("Profit", "sum")  
    )  
)  
  
region_analysis
```

Out[33]:

	Revenue	Profit
sales_region		
Canada West	107674.34	64341.00
Central West	9324.94	5569.91
Mexico Central	330362.03	197072.61
Mexico South	87253.65	51991.14
Mexico West	61299.98	36623.57
North West	847826.72	505783.29
South West	320804.78	191437.26

```
In [34]: plt.figure(figsize=(10,6))  
  
plt.scatter(  
    region_analysis["Revenue"],  
    region_analysis["Profit"],  
    s=150  
)  
  
for region in region_analysis.index:  
    plt.text(  
        region_analysis.loc[region, "Revenue"],  
        region_analysis.loc[region, "Profit"],  
        region  
    )
```

```
plt.xlabel("Revenue")  
plt.ylabel("Profit")  
plt.title("Revenue vs Profit by Region")  
plt.show()
```



```
In [35]: # Store Ranking  
store_analysis = (
```

```
master.groupby("store_name")
    .agg(
        Revenue=("Sales", "sum"),
        Profit=("Profit", "sum")
    )
    .sort_values("Revenue", ascending=False)
)

store_analysis.head(10)
```

Out[35]:

	Revenue	Profit
store_name		
Store 13	170398.94	101714.65
Store 17	157695.90	94087.96
Store 15	115981.46	69148.47
Store 11	115334.62	68857.27
Store 24	111378.46	66459.56
Store 7	111311.54	66424.49
Store 3	110877.88	66055.01
Store 16	110485.16	65911.35
Store 6	98114.78	58553.21
Store 12	87623.61	52257.30

```
In [36]: # Customer Segmentation
customer_analysis = (
    master.groupby("customer_id")
        .agg(
            Revenue=("Sales", "sum"),
            Orders=("transaction_date", "count")
        )
)
```

```
customer_analysis["Segment"] = "Bronze"

customer_analysis.loc[
    customer_analysis["Revenue"] >= 5000,
    "Segment"
] = "Gold"

customer_analysis.loc[
    (customer_analysis["Revenue"] >= 2500) &
    (customer_analysis["Revenue"] < 5000),
    "Segment"
] = "Silver"

customer_analysis.head()
```

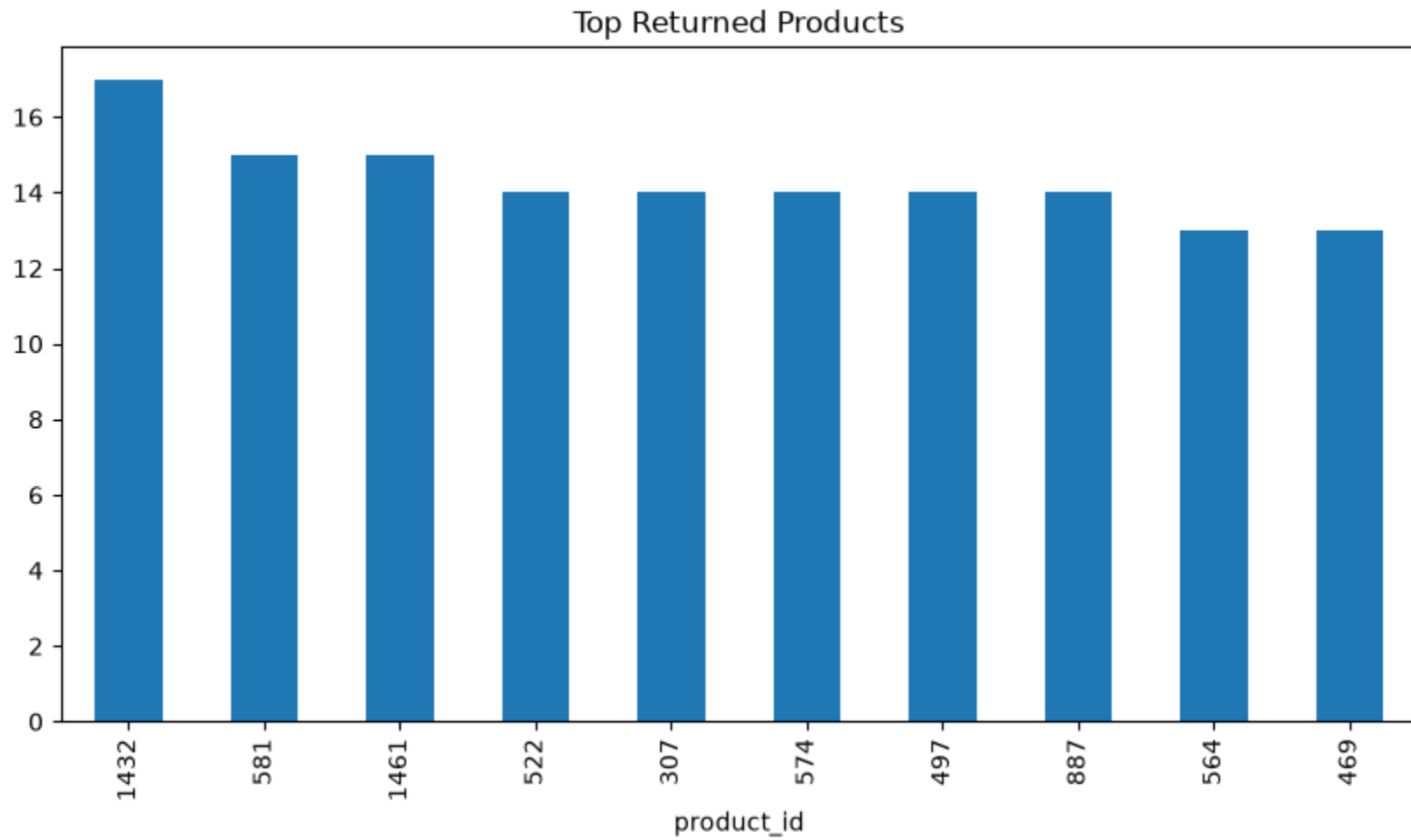
Out[36]:

	Revenue	Orders	Segment
customer_id			
3	203.17	30	Bronze
5	1.08	1	Bronze
6	147.71	16	Bronze
8	259.14	42	Bronze
9	89.58	13	Bronze

```
In [37]: # Returns Analysis
top_returns = (
    returns.groupby("product_id")["quantity"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

top_returns.plot(
    kind="bar",
    figsize=(10,5)
)
```

```
plt.title("Top Returned Products")  
plt.show()
```



In []: